

Anatomy of the Long Tail

Sharad Goel

Microeconomics and Social Systems, Yahoo! Research

Joint work with Andrei Broder, Evgeniy Gabrilovich, and Bo Pang

Infinite Selection

In the last 15 years, [infinite-inventory](#) stores have become commonplace:

- Amazon ($\approx$ 5 million books)
- Netflix ($\approx$ 100,000 movies)
- iTunes ($\approx$ 10 million songs)

Such stores offer orders of magnitude more selection than traditional bricks-and-mortar retailers:

- Barnes & Noble ($\approx$ 100,000 books)
- Blockbuster ($\approx$ 3,000 movies)
- WalMart ($\approx$ 50,000 songs)

Long Tail Markets

The **long tail markets** of infinite-inventory stores exhibit two near-universal properties:

- 1 The vast majority of products are “misses”
- 2 These “worst-sellers” in aggregate account for a large fraction of sales
25% of Netflix sales and 30% of Amazon sales
are for products not available in traditional stores

Chris Anderson, “The Long Tail”

“If you combine enough of the non-hits, you’ve actually established a market that rivals the hits.”

First vs. Second Order Effects

Most past work has focused on the **direct effects** of larger inventory (i.e., the percentage of sales accounted for by tail inventory)

[Anderson, Brynjolfsson et al., Elberse et al., Tan et al.]

We instead focus on the **indirect effects** of larger inventory (i.e., the effect tail inventory has on attracting users, indirectly driving sales of other items)

Myopic (item-centric) vs. **systemic** (user-centric) analysis

The Impact of Inventory

Definitively assessing the impact of large inventories is hard due to a variety of confounding factors. For example:

- Netflix does not have late fees
- Amazon typically undercuts the prices of its competitors
- iTunes has instant downloads

We instead ask a simpler, but related question:

How many people consume tail content?

- If not very many, then indirect effects of the tail are likely small
- If there are a lot, then potentially large effects

How Many People Consume Tail Content?

Two Views of the World

① Most people are sheep

They only watch Avatar, listen to Beyoncé, and read Harry Potter
Most people are perfectly happy with limited inventory

② Most people are a bit eccentric

With some regularity, most people consume content not available in bricks-and-mortar stores
Most people are affected by greater product selection

Ordinary People With Extraordinary Tastes

Looking at movies, music, web search, and web browsing we find that **nearly everyone is at least a bit eccentric** (i.e., nearly everyone consumes tail content). For example:

- 90% of Netflix users and 95% of Yahoo! Music users have consumed items not available in large, bricks-and-mortar retailers
- 35% of Netflix users and 70% of Yahoo! Music users regularly consume tail items

A Hypothetical Example

Suppose we're trying to decide whether Yahoo! Search should support academic queries (e.g., similar to Google Scholar)

- **Myopic Analysis**

Predict academic search would generate 100K queries per month

Determine that such queries don't monetize well

Conclude the cost of implementation outweighs projected revenue

- **Systemic Analysis**

Predict academic queries would attract 10K more users

Determine these users would each issue 100 queries per month
(many of which would monetize well)

Conclude the 1M increase in query volume justifies costs

Data

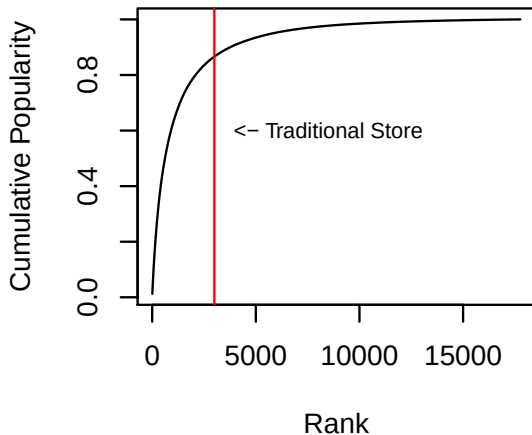
	Items	Users	Observations
Movies	20K	400K	100M
Music	700K	2M	800M
Search – Queries	500M	60M	3B
Search – Clicked Domains	20M	60M	2B
Web Browsing	2M	100K	300M

Search and music data from Yahoo!

Movie data from Netflix

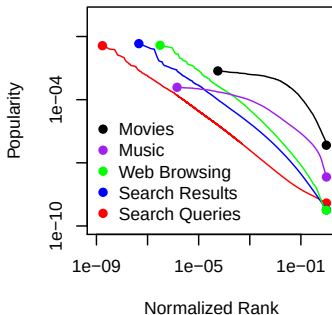
Web browsing data from Nielsen

The Long Tail of Netflix



1 - 100: 15% consumption
3,000 - ∞ : 15% consumption

Heavy Heads & Long Tails



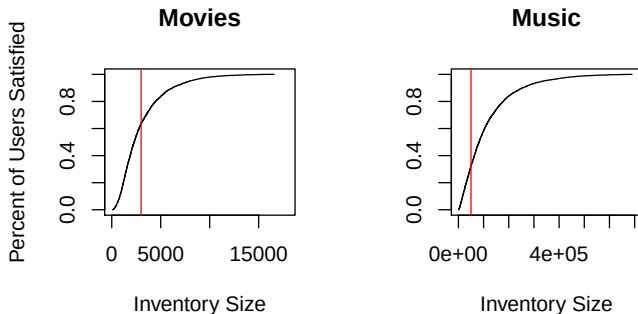
Music

1 - 1000: 15% consumption
50,000 - ∞ : 35% consumption

Web Browsing

1 - 10: 15% consumption
10,000 - ∞ : 20% consumption

Consumer Satisfaction

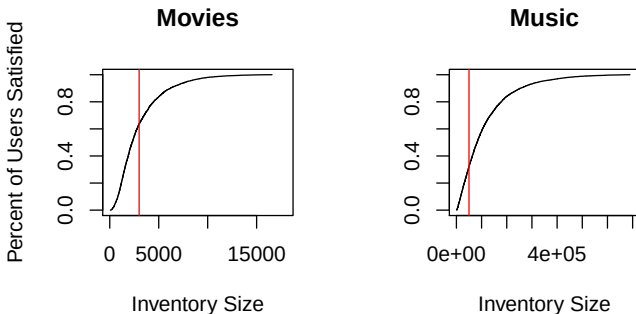


Definition

A user is **satisfied** by an inventory of size k if at least 90% of the items they consume are in that inventory.

35% of Netflix users and 70% of Yahoo! Music users are not satisfied by traditionally-sized inventories

First vs. Second Order Effects

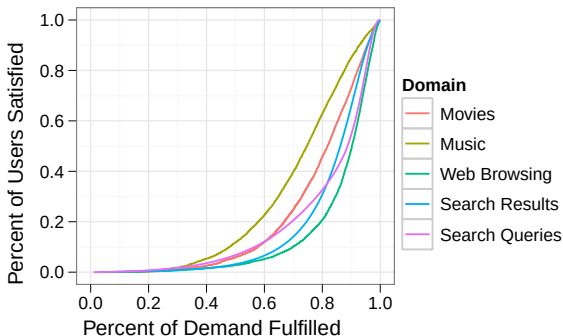


Movie Inventory: 3,000 $\rightarrow$ 3,500

First-Order Effect: 2% increase in consumption

Second-Order Effect: 7% increase in satisfaction

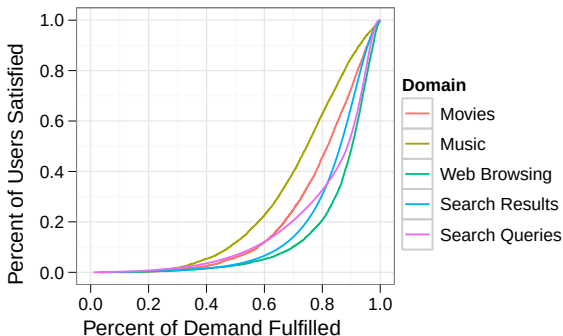
First vs. Second Order Effects



First-Order Effect: Percent of Demand Fulfilled (x-axis)

Second-Order Effect: Percent of Users Satisfied (y-axis)

First vs. Second Order Effects



Web Search

First-Order Effect: 1% increase in consumption (95% $\rightarrow$ 96%)
Second-Order Effect: 6% increase in satisfaction (80% $\rightarrow$ 86%)

Google Books

Google has digitized and indexed millions of books, largely without explicit consent from copyright holders.

The project has provoked the ire of authors, publishers, and legal scholars who claim:

- massive copyright infringements that violate fair use
- monopoly issues and the opportunity for price gouging
- little protection of user privacy

Criticisms focus on the impact of Google Books on the book industry

Google Books

The Open Book Alliance points out that Google Books has serious implications for web search as well.

In an amicus brief filed in opposition to the proposed Google book settlement:

“Digital rights to virtually all out-of-print books provide Google with a decisive advantage in responding to tail queries.”

Are Niche Products Appreciated?

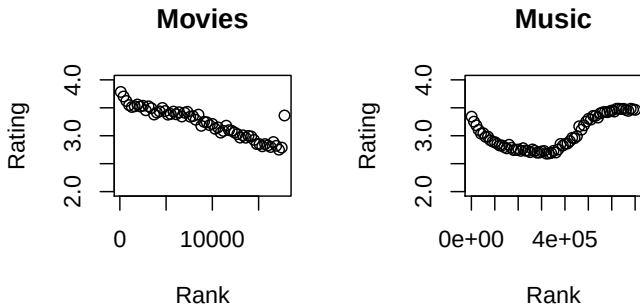
Study of Quickflix DVD rental data shows tail items received below average user ratings [Elberse]

Obscure products thought to be in a state of double jeopardy [McPhee]:

- Consumers unlikely to find obscure products
- Those that do find such products, don't appreciate them

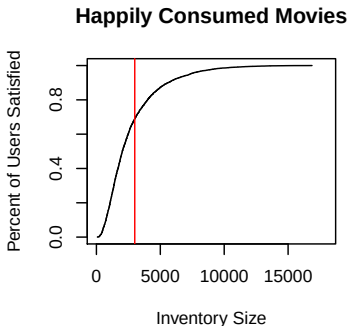
Previous Conclusion: Since people don't really like niche products, tail inventory is not that important

Are Niche Products Appreciated?



- Tail movies receive lowest ratings (consistent with previous work)
- Tail songs receive highest ratings

Are Niche Products Appreciated?



Only consider instances where a user rates a movie highly (4 out of 5 stars)

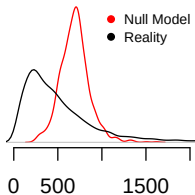
30% of Netflix users and 55% of Yahoo! Music users not satisfied by traditionally-sized inventories (compared to 35% and 70% when ignoring ratings)

A Null Model of Consumer Behavior

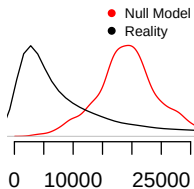
- 1 User randomly decides how many items to consume
Preserves the empirical, domain-specific distribution
- 2 Randomly selects items proportional to popularity
Does not have any other intrinsic preference for the head or tail

A Null Model of Consumer Behavior

Movies



Music

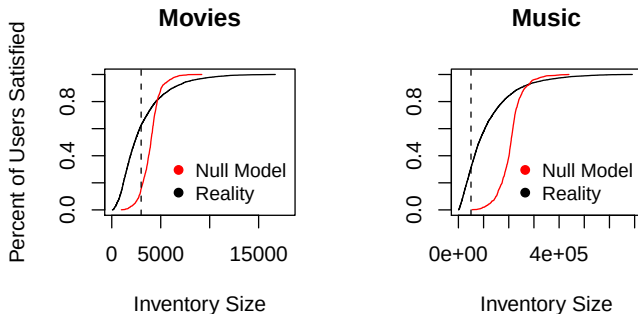


Definition

A user's **eccentricity** is median rank of items it consumed

Users much less eccentric in reality than under null model

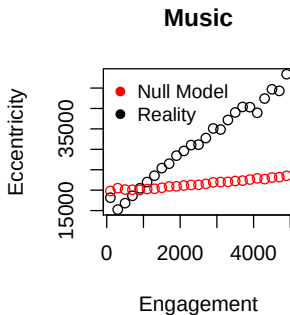
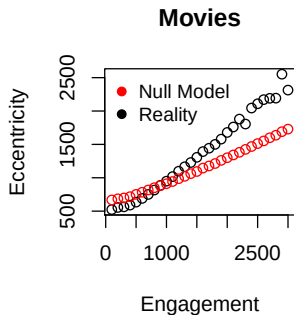
A Null Model of Consumer Behavior



Users much harder to satisfy under null model than in reality

15% of movie-model users and 0% of music-model users satisfied by traditionally-sized inventories (compared to 65% and 30% in reality)

Heavy vs. Light Users



Theoretical Analysis

Sticky Model of Consumer Behavior

- One consumer, two retailers
- Consumer randomly selects a product it would like to buy
- Consumer picks a retailer, and checks if retailer carries that product
 - If product is available, consumer buys it
 - If not, consumer tries to buy the product at the other retailer
- Consumer starts search at retailer from whom it most recently purchased an item

Sticky Model

If retailer A 's inventory is a superset of retailer B 's inventory, then A will eventually attract all of the user's business

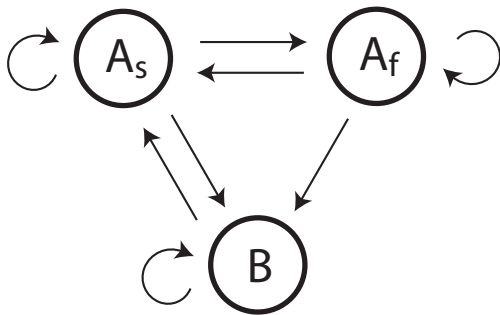
Small differences in inventory may lead to large revenue disparities due to repeat patronage (winner-take-all)

Theorem

*More generally, market share depends on the fraction of **unique** inventory carried by each retailer*

In the sticky model, **differentiation** is key

Sticky Model — Proof



$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n I_{A_s, k} = Q(I_A) Q(I_A | I_A \Delta I_B) + Q(I_A \cap \overline{I_B}) Q(I_B | I_A \Delta I_B)$$

Shared Inventory Model

We alter the sticky model to resemble Amazon Marketplace

- Two competing retailers
- One retailer (the marketplace) makes available the inventory of other sellers
- If marketplace retailer doesn't have a product, it directs user to affiliates

Ordinarily, the marketplace (e.g., Amazon) receives a commission from third-party transactions.

Our model excludes commissions, isolating the second-order effect of consumer satisfaction

Shared Inventory Model

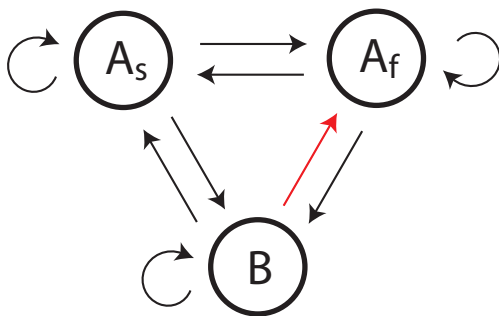
If retailer A and its affiliates together offer a superset of retailer B 's inventory, then A will eventually attract all of the user's business.

Theorem

In terms of second-order effects (i.e., repeat patronage), the marketplace retailer can effectively treat its affiliates' inventory as its own

In the shared inventory model, affiliates help make you a **starting point**.

Shared Inventory Model — Proof



$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n I_{A_s, k} = Q(I_A) Q(I_A \cup I_C | (I_A \cup I_C) \Delta I_B) + Q(I_A \cap \overline{I_B}) Q(I_B | (I_A \cup I_C) \Delta I_B)$$

Summary

Across a variety of domains, we find that a substantial fraction of users regularly consume niche content

This suggests that the benefit of tail inventory extends beyond direct revenue to second-order gains associated with increased consumer satisfaction and repeat patronage

When deciding whether or not to stock the latest worst-sellers, the pertinent question is not, *How many copies will I sell?* but rather, *How many customers will I gain?*

Some Caveats

- Possible selection bias

Netflix users may be precisely those individuals demanding infinite selection

- Recommender systems may inflate eccentricity

Even if typical individuals are consuming—and appreciating—niche items, they may not miss the absence of such selection.

These concerns are mitigated by the consistency of our findings across several diverse domains—movies, music, web search, and web browsing.

In web search and web browsing, the effects of selection bias and recommender systems seem minimal.

Another Caveat

Inventory size is just one factor among many that affects customer satisfaction

Consider [Redbox](#)—a DVD vending company that rents movies via kiosks that offer 100–200 titles

DVD vending companies control about 20% of the DVD rental market



Blog Post + Paper:

messymatters.com/longtail

